

The $\cos 2\phi$ modulation on tensor-polarized ${}^6\text{Li}$: projected inclusive and coherent measurements

2026-08-17 · development run 5 · polligen (evgen/) + polli_fastsim (fastsim/) · companions: plans/06 (background budget), docs/note_cos2phi_coherent_6Li.md, reports/polarized_li_primer.html (pedagogical introduction) · external numbers verified against primary sources; this revision adds dedicated fact checks of the moment constraint and of the far-forward detection chain

Two projected EIC measurements of the $\cos 2\phi$ azimuthal modulation on a transversely tensor-polarized ${}^6\text{Li}$ beam: inclusive DIS with only the scattered electron detected, and coherent diffraction with the ${}^6\text{Li}$ detected intact in its 1^+ ground state. The modulation is driven by photon double helicity flip — the gluon-transversity sector, closed to quarks and to bound nucleons [1,2] — and has never been measured on any target. This note gives the projections, the assumptions behind each number, and the checks made on both.

13.5%

coherent-tag acceptance at IP6 with the 0.20 GeV envelope cut, $\exp(-B \cdot p_T^2 / \text{cut})$, $B = 50 \text{ GeV}^{-2}$ [9–20% for $B \in 40\text{--}60$]. The cut is documented for top-rigidity beams; for ${}^6\text{Li}$ it is an optics assumption (§4.1).

$1.1 \times 10^7 / 10^8$

tagged coherent events in 1 / 10 EIC years (scenario $f_0 = 0.04$); best-bin $\delta A = 1.9 \times 10^{-3} / 6 \times 10^{-4}$, i.e. 5 σ floors at 1.0% / 0.3% modulation

0.036

deformation-anchored $\langle a_2 \rangle$ of the tagged sample at $P_{zz} = 0.6$ [band 0.018–0.059; one-sided $\times 0.73$ rate-weighting systematic], scaled from the polarized-deuteron CGC calculation [13]; sign flip vs the deuteron predicted

1 · The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle θ_m with the beam, the inclusive cross section per spin projection m is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

The $\cos 2\phi$ term requires the virtual photon to flip helicity by two units. No spin-1/2 constituent can absorb that — no quark, and no bound nucleon — so $\Delta(x, Q^2)$ is purely gluonic at leading twist and vanishes for any nucleus that is only protons and neutrons [1]. At leading order it is generated by the gluon transversity distribution δG through an $O(\alpha_s)$ photon–gluon convolution,

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left(\sum_q e_q^2 \right) x^2 \int_x^1 \frac{dy}{y^3} \delta G(y, Q^2)$$

(Detmold–Shanahan sign convention [25]; conventions differ between papers). Δ has never been measured for any target; the only prior experimental document is a JLab letter of intent without a sensitivity projection [3]. Magnitude guidance is thin. The single moment estimate is the Sather–Schmidt bag-model calculation for the spin-3/2 Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$ [4], which sets a per-mille scale for the asymmetries; percent-level numbers exist only for maximal-transversity ($\Delta_{Tg} = \Delta g$) proton–deuteron Drell–Yan scenarios that their authors call optimistic [5]. We therefore work with a scenario band $\Delta/F_1 \in 10^{-3}\text{--}10^{-2}$.

The measured amplitude and its inversion back to the structure function are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = \hat{A} \frac{y^2 D_\phi}{1-y}$$

Approximations in the fastsim set: $A_{//} = D \cdot g_1 / F_1$ (γ^2 -suppressed term dropped), $g_2 = g_2^{\text{WW}}$, no target-mass corrections. The last matters only at high x and low Q^2 ; the sweet-spot bins sit at $x \leq 0.06$ where $\gamma^2 = 4M^2 x^2 / Q^2 < 10^{-2}$, so it is a labeling caveat rather than a bias. Structure functions are TOY parameterizations with PDF-grid

(CT18NLO/NNPDFpol11) backends behind the same interface: rates are good to a factor ~ 1.5 , and asymmetry shapes are scenarios by construction.

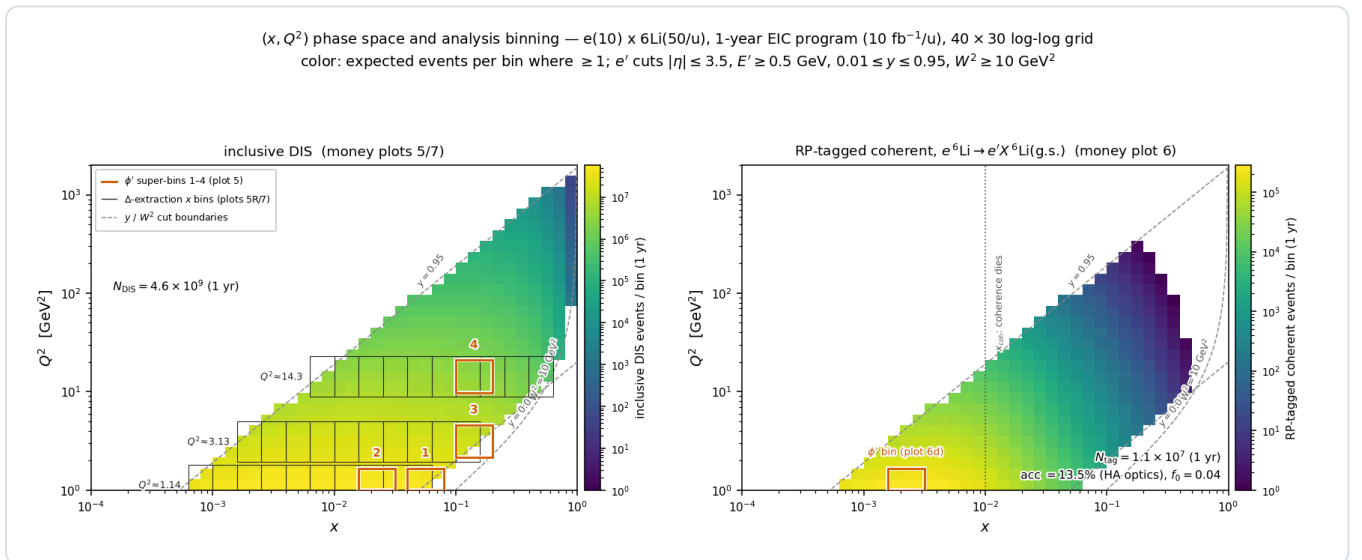
2 • Machinery, statistics, and binning

Generator. No public generator produces polarized e+A events for $A > 1$ (BeAGLE is unpolarized; DJANGO/PEPSI are nucleon-level and longitudinal-only), so polligen implements the missing piece — the spin-density matrix $\otimes$ structure-function master formula with spin-correlated sampling — and reuses everything else. Validation gates: bin-wise identity with the analytic asymmetry formulas at rtol 10^{-12} , pseudo-experiment spreads closing on every analytic error map, and unbiased $\cos 2\phi$ recovery under ϕ -acceptance holes.

Full-luminosity statistics. For a binned estimator, drawing each ϕ -bin count from a Poisson distribution with the exact expected yield is statistically identical to event-level sampling, so a full-luminosity pseudo-experiment costs 24 Poisson draws (`sample.phi_histogram_pseudo`). Event-level sampling remains available and is what the closure tests use.

Estimator. The amplitude comes from a binned least-squares fit of $N(\phi') = C(1 + A \cos 2\phi')$, not the $2\langle \cos 2\phi' \rangle$ moment: with realistic ϕ holes the moment is biased by more than the signal, while the fit stays unbiased. Errors follow $\delta A = \sqrt{(2/N)/P_{ZZ}}$, validated by pseudo-experiments. The fit carries no $\cos \phi'$ term; the m-symmetric tensor fills set $a_1 = 0$ by construction.

Binning. The four displayed (x, Q^2) super-bins are the top significance picks per Q^2 band (the raw map clusters at the lowest Q^2 ; the Q^2 lever arm is kept because it is the consistency check against power-suppressed contributions, §6). Each super-bin merges 3×3 cells of the 40×30 analysis grid. Rates use per-nucleon luminosity, F_2^A/A , scattered-electron cuts $|\eta| \leq 3.5$, $E' \geq 0.5$ GeV, $0.01 \leq y \leq 0.95$, $W^2 \geq 10$ GeV², and the whole luminosity in the transverse-tensor fill; projections are quoted for the 1-year ($10 \text{ fb}^{-1}/u$) and 10-year ($100 \text{ fb}^{-1}/u$) programs.



Where the events are and how they are binned ($e^{10} \times {}^6\text{Li}50/u$, 1-year program, 40×30 log-log grid; color = expected events per bin where ≥ 1 ; dashes: y and W^2 cut boundaries). **Left — inclusive DIS** (4.6×10^9 accepted events in year 1): boxes 1–4 are the ϕ' super-bins of money plot 5; gray combs are the three Q^2 slices ($Q^2 \times [1/1.6, 1.6]$) whose merged x-bin pairs become the data points of the amplitude-vs-x panel and the Δ extraction (plots 5R/7), kept where $N \geq 10^3$ and $\delta A \leq 8 \times 10^{-3}$. **Right — RP-tagged coherent channel:** the same map $\times f_{\text{coh}}(x) \times$ the 13.5% tag acceptance. $N_{\text{tag}} = 1.1 \times 10^7$ in year 1; 98% of the tagged sample sits at $x < 0.03$ (median 3.5×10^{-3}). The box is the super-bin of money plot 6(d); plot 6(c) is this map summed over Q^2 .

3 • The Δ model and the moment constraint

What the constraint is. Sather and Schmidt computed the lowest moment of Δ in the MIT bag model for the spin-3/2 Δ^{++} baryon and its Q^2 evolution; they give no x-shape [4]. The explicit $\alpha_s(Q^2)$ is the leading-order

Wilson coefficient of the photon–gluon box, so the $O(\alpha_s)$ suppression relative to F_1 is model-independent; the -0.012 is the bag estimate. Lattice QCD adds two data points, with heavy caveats: the bare ϕ -meson matrix element $A_2 = 0.23(2)(5)$ at $m_\pi = 450$ MeV [25] converts to $\int x \Delta|_\phi \approx -0.006 \alpha_s$ (our conversion, unrenormalized), while the NPLQCD deuteron gives a δG moment of $-0.010(3)$, bare at $m_\pi \approx 806$ MeV [27] — a transversity-to-momentum ratio roughly ten times below the ϕ 's, and implying the opposite sign for the deuteron's Δ moment. No calculation of Δ exists for ${}^6\text{Li}$ or any $A > 2$ nucleus; the only nuclear model is a $\Delta\Delta$ -isobar estimate for the deuteron [34], and the literature brackets nuclear Δ in both directions — suppressed as in the NPLQCD deuteron, or enhanced by binding like the EMC effect, as the polarized-ion white paper argues [50].

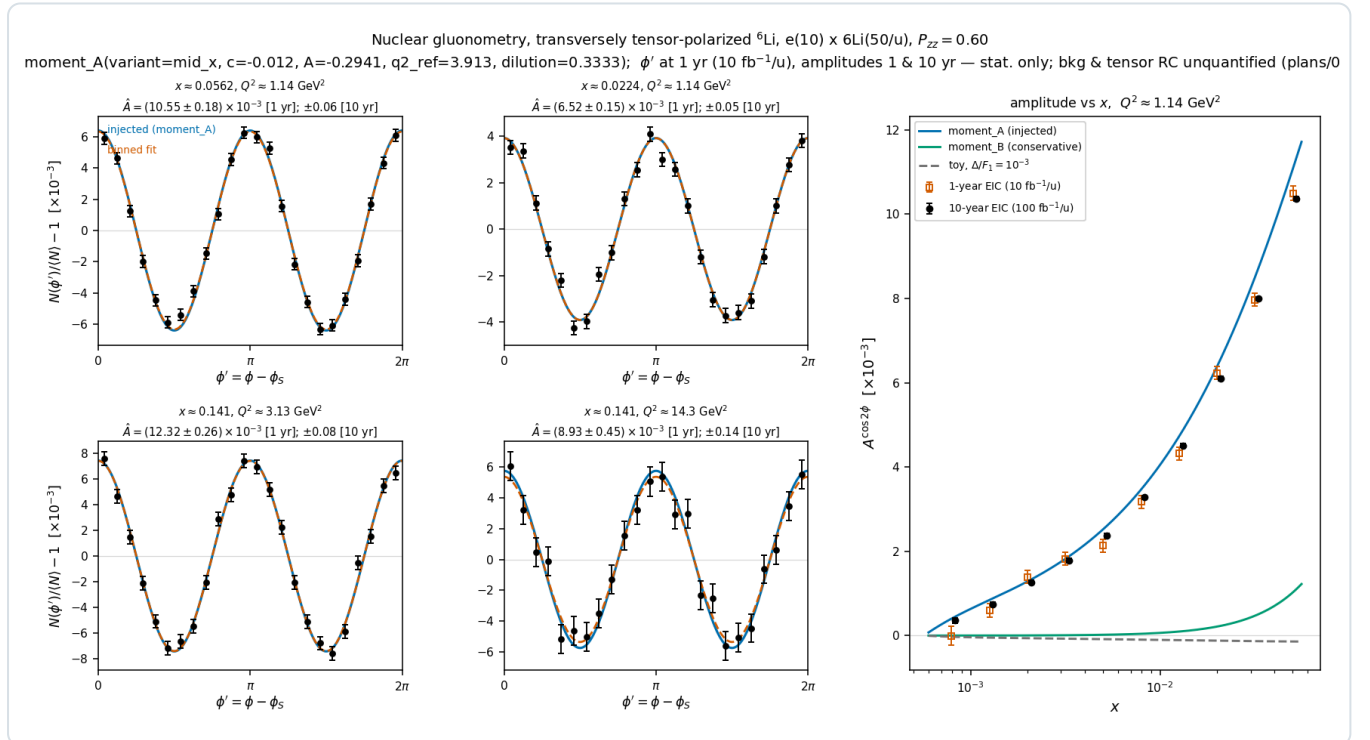
What we do with it. All Δ models live in one registry (`polli_fastsim/delta_models.py`). The production default, `moment_A`, is an ansatz normalized to the bag moment applied to the per-nucleon Δ of ${}^6\text{Li}$:

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.294$$

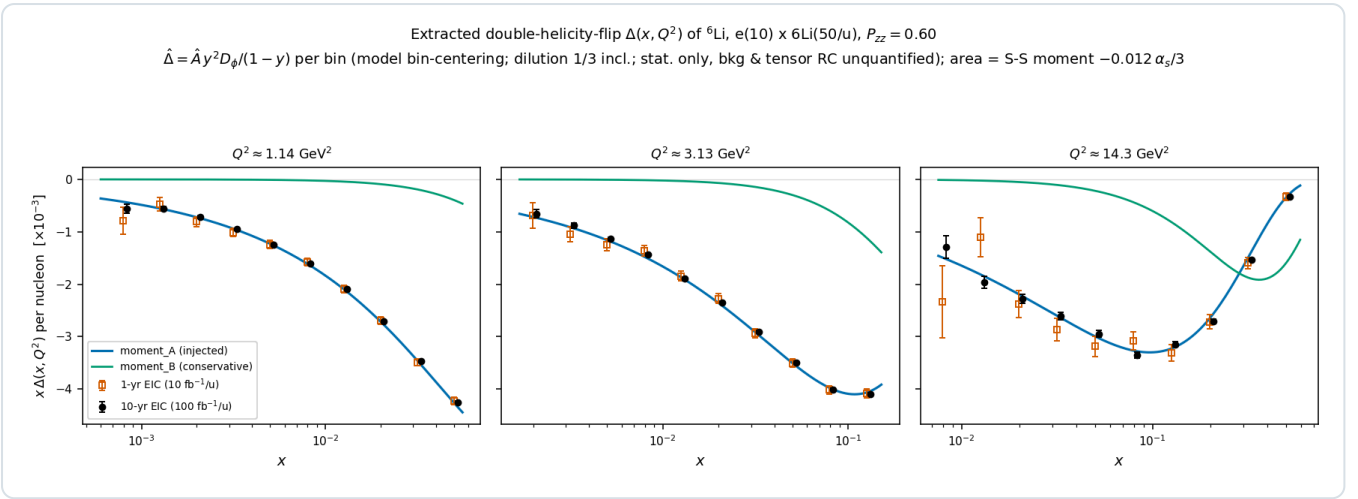
with $(a, b) = (0.7, 3)$ and the $1/3$ restricting the exotic glue to the polarized deuteron-like pair (the whole-nucleus P_{ZZ} stays a separate factor, so the two never double-count). The constraint is imposed at the rate-weighted $\langle Q^2 \rangle = 3.9$ GeV² of the accepted phase space; because F_1 evolves, the realized moment drifts between -4% and $+7\%$ across the plotted Q^2 slices. α_s comes from the CT18NLO table (the LO analytic fallback runs 14% high). The solved $A = -0.294$ matches the EPPS21-based `money_delta` value -0.31 at its $\langle Q^2 \rangle = 7.4$ GeV².

`moment_B` applies the same constraint without the F_1 factor and lands a factor 2–8 lower; the A-vs-B spread is the dominant ansatz systematic, and both curves are drawn on every plot.

What it is not. Porting a single-hadron bag moment to a per-nucleon nuclear structure function has no literature precedent; both the normalization and the x-shape are labeled scenario choices, bracketed by `moment_B` below and by the possible nuclear enhancement above. What the projection then shows is that the measurement has the precision to decide: with the moment constraint in place, the x-shape separates interpretation A from B bin by bin.



Money plot 5 — inclusive gluonometry ($e10 \times {}^6\text{Li}50$, $P_{ZZ} = 0.60$, Δ : `moment_A`, dilution 1/3). Left 2×2: ϕ' pseudo-data with 1-year statistical error bars in the four sweet-spot super-bins ($Q^2 = 1.1 \rightarrow 14$ GeV², $x \approx 0.02\text{--}0.14$); injected curve (blue), binned fit (vermillion); annotations give δA for both programs. Right: extracted amplitude vs x with 1-year (open) and 10-year (filled) error bars against the `moment_A`, `moment_B`, and `toy-10-3` curves. Amplitudes are $(0.7\text{--}1.2) \times 10^{-2}$ at the sweet spots against $\delta A = (1.5\text{--}4.5) \times 10^{-4}$ in year 1 — every bin $\geq 21\sigma$, so the A-vs-B spread is resolved by the x-shape. Detection needs only the scattered electron.



Money plot 7 — the extracted $\Delta(x, Q^2)$. The amplitudes of money plot 5 unfolded to the structure function (model bin-centering, $\delta\Delta = \delta A \cdot |\partial\Delta/\partial A|$), shown as $x\Delta$ per nucleon at the three sweet-spot Q^2 slices with independent 1-year (open) and 10-year (filled) pseudo-measurements against the moment_A and moment_B curves; the area under each curve is the ported bag moment ($\div 3$ dilution). Best bins reach $\delta\Delta \approx (0.7-1.4) \times 10^{-3}$ in year 1 on $\Delta \approx -0.03 \dots -0.09$.

4 • The coherent channel

4.1 • Detecting the intact ${}^6\text{Li}$ at ePIC

In coherent diffraction, $e \, {}^6\text{Li} \rightarrow e' \, X \, {}^6\text{Li}(\text{g.s.})$, the recoil keeps the beam's magnetic rigidity exactly ($A/Z = 2$; the diffractive momentum loss $x_p < 0.01$ changes it by $< 1\%$) and emerges at $\theta = p_T/p \lesssim 1.7$ mrad. Everything below 5 mrad enters the far-forward system: Bo spectrometer (5.5–20 mrad, charged tracks and photons above ~ 50 MeV), Off-Momentum Detectors (~ 30 – 60% of beam rigidity), Roman Pots at $z \approx 26/28$ m ($\theta < 5$ mrad, ~ 60 – 95% rigidity for off-rigidity tracks, AC-LGAD sensors with $\sim 140 \, \mu\text{m}$ and ~ 30 ps resolution), and the ZDC for neutrals [7,8,9]. A beam-rigidity recoil rides inside the beam envelope and is invisible to all of them except the Roman Pots, and there only above the 10σ beam-envelope cut — at low $|t|$ the intact recoil cannot be tagged at IP6 at all [10,51]. With an exponential coherent t -distribution the tag acceptance is analytic:

$$\epsilon_{\text{tag}} = e^{-B p_{T,\text{cut}}^2} = 13.5\% \quad (p_{T,\text{cut}} = 0.20 \, \text{GeV}, B = 50 \, \text{GeV}^{-2}); \quad 4 \times 10^{-5} \quad (0.45 \, \text{GeV})$$

Where the cuts come from. The 0.20 GeV floor is the documented EIC requirement for high-acceptance (low-divergence) optics at top rigidity — divergence $\leq 73 \, \mu\text{rad}$ for 275 GeV protons [7,48] — and matches the one documented light-ion number, coherent ${}^3\text{He}$ tagging limited to $p_T > 0.20$ GeV at IR-6 [10]. The 0.45 GeV value is our derivation (150 μrad high-divergence proton optics applied to the 300 GeV ${}^6\text{Li}$ momentum), not a documented number. No beam-optics parameters are published for any lithium beam; if the ${}^6\text{Li}$ beam ran with proton-class divergence at this rigidity, the 10σ cut would land at 0.5–0.7 GeV and close the IP6 channel entirely. The cut is the single most consequential machine input to this measurement (§6, #1).

The IP6 fallback and the IR-8 fix. Standard EIC practice at IP6 is to establish coherence by vetoing breakup rather than tagging the recoil: dissociation protons (rigidity 0.50) hit the Off-Momentum Detectors, neutrons the ZDC, de-excitation photons the Bo calorimeter and ZDC, and ${}^3\text{He}$ -like fragments (0.75) the Roman-Pot window [51]. Direct low- $|t|$ tagging is the flagship argument for the second interaction region: its lattice adds a secondary beam focus ~ 45 m downstream where the beam is $\sigma \approx 0.15 \times 0.1$ mm, so Roman Pots at $z = 44.0/45.5$ m reach $p_T \approx 0$. The published intact-recoil efficiencies there are d 47%, ${}^3\text{He}$ 32%, ${}^4\text{He}$ 29%, ${}^7\text{Li}$ 17.8% at top energy (rising steeply at lower energies — ${}^3\text{He}$ reaches 99.8% at 5×41) [10]; no ${}^6\text{Li}$ case has been published, and our interpolation is $\approx 20\%$. One degeneracy remains even there: a Roman-Pot hit measures only A/Z , so intact ${}^6\text{Li}$, α , and d at the same velocity are identical in trajectory and timing; the second-detector design addresses this with a proposed Z^2 -scaling Cherenkov behind the pots (R&D status) [8], while for the IP6 AC-LGADs no charge-discrimination study exists — signal amplitude is used for position interpolation only, and even the

minimum-angle ion acceptance there is not yet fully established [49]. That question (open item #19) decides whether IP6 coherent-tag purity is event-by-event or statistical (§5).

4.2 • Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 \text{ [0.02–0.08]}$$

No light-nucleus diffractive prediction exists (the lightest published is Ca), so the f_0 band brackets HERA ep diffraction (10–15% of DIS [44]) and heavy-A saturation estimates (20–25% [45]). $x_{\text{coh}} = 0.01$ is where the coherence length $l_c = 0.105 \text{ fm}/x_p$ drops below the ${}^6\text{Li}$ size. The slope $B = 50 \text{ GeV}^{-2}$ [40–60] follows from $B = \langle r^2 \rangle / 3$ with the matter radius (the charge radius gives ≈ 57 [47]; gluonic distributions may be wider [42]); the exponential form is valid below the form-factor minimum at $|t| \approx 0.31 \text{ GeV}^2$ [11,12]. Writing f_{coh} against Bjorken x rather than x_p is optimistic for high-mass diffraction; the f_0 band covers it. Yields: 1.1×10^7 tagged events in year 1, 1.1×10^8 over ten; best super-bin $\delta A = 1.9 \times 10^{-3} / 6 \times 10^{-4}$.

4.3 • The anchored amplitude

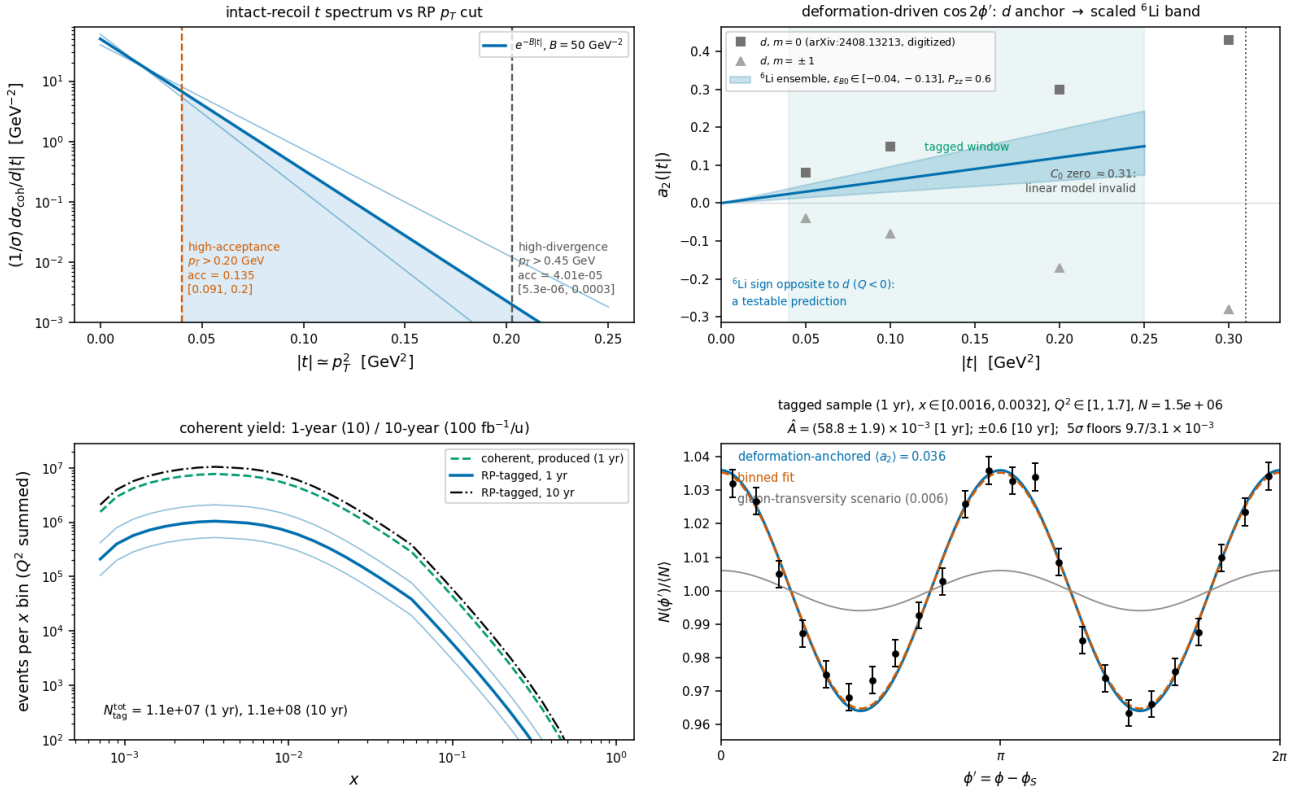
The $\cos 2\phi'$ modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron CGC calculation of Mäntysaari, Salazar, Schenke, Shen, and Zhao [13] — the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's follow-ups are unpolarized [14,15]). Their digitized coefficients (`coherent.MANTYSAARI_A2_DEUTERON`) show a_2 growing linearly in $|t|$ with $a_2(m=0) = -2a_2(m=\pm 1)$, from the 6% D-wave modulating the diffractive slope. For a fill with populations p_m the ensemble coefficient is proportional to P_{zz} :

$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \text{ [0.018–0.059]}$$

$\varepsilon_{B0} \equiv \Delta B_0/B$ is scaled from the deuteron's 0.21 by two dials that disagree: the relative quadrupole deformation $Q/\langle r^2 \rangle$ of ${}^6\text{Li}$ is about 1/5 of the deuteron's and opposite in sign, while the α -d asymptotic D/S ratio is disputed — the transfer measurement gives $\eta = +0.0003(9)$, consistent with zero [16], where older sub-Coulomb analyses gave $\eta \approx -0.01$. We take $\varepsilon_{B0} \in -(0.04-0.13)$, default -0.08 , with near-zero deformation open (the 5σ floors cover its detection or exclusion). The band is a physics statement: even exact few-body theory misses the ${}^6\text{Li}$ quadrupole cancellation by $\sim 3\times$ [17], and the gluonic deformation need not cancel like the charge one. The **sign flip relative to the deuteron is a testable prediction** ($Q({}^6\text{Li}) < 0$). Existence proofs at this $|t|$: e-d $T_{20} = -(0.1-0.3)$ [19]; HERMES $A_{zz}(d) \approx -1\%$ at $x \approx 0.01$ [20], coherent-double-scattering in origin [21–23]. The flat gluon-transversity term is separately bounded at $3 \times 10^{-3} - 10^{-2}$, from the bare lattice ϕ transversity [25] divided by the $\sim 10\times$ NPLQCD deuteron suppression [27] (both at unphysical pion masses), with the maximal Drell–Yan scenario [5] as the ceiling. The equal-rate population average behind $\langle a_2 \rangle$ carries a one-sided $\times 0.73$ rate-weighting systematic (audited; inside the ε_{B0} band).

What the tagged sample measures: the two mechanisms separate by shape. Deformation is t -linear, sign-flipped vs the deuteron, and should weaken toward small x_P (JIMWLK evolution washes out deformation [15]); exotic glue is flat in t and survives at small x , where it connects to the elliptic-gluon program [6]. A two-component fit $a_2(t) = c_{\text{def}} \cdot |t| + a_g$ per x_P bin has both components $\geq 5\sigma$ above the statistical floor at $100 \text{ fb}^{-1}/u$. Incoherent breakup is m -state-blind in this mechanism [13]: it dilutes but cannot fake the modulation.

Coherent $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$, transversely tensor-polarized, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
 nario rates; modulation anchored on the polarized- d CGC calculation, PLB 858:139053 -- plans/06 SS6.4b; statistical only, backgrounds/purity and tensor RC unquanti



Money plot 6 — coherent channel, $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$ (same beam, $P_{zz} = 0.60$). Top left: recoil t -spectrum vs both optics' p_T cuts with the slope band. Top right: digitized deuteron a_2 per m -state [13] vs the scaled ^6Li ensemble band (sign flip predicted); tagged window and form-factor-zero exclusion marked. Bottom left: tagged yield vs x for both programs with the f_0 band ($N_{\text{tag}} \approx 1.1 \times 10^7 / 1.1 \times 10^8$). Bottom right: ϕ' pseudo-data (1-year statistics) at the anchored $\langle a_2 \rangle = 0.036$ with the flat gluon-transversity scenario (gray); extraction $\hat{A} \pm 1.9 \times 10^{-3}$ (1 yr), $\pm 0.6 \times 10^{-3}$ (10 yr).

5 • Backgrounds for the coherent tag (full budget: plans/06)

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
$\alpha + d$ breakup (threshold 1.474 MeV; the 2.186 MeV 3^+ state, $\Gamma = 24 \text{ keV}$, decays $\sim 100\%$ $\alpha + d$ [39])	Both fragments at $R = 1.000$, same trajectory and velocity as intact ^6Li : tracking and ToF blind; only dE/dx ($\propto Z^2: 9/4/1$) separates	RP charge discrimination — open question #19 : no study exists for the IP6 AC-LGADs (amplitude is used for position; the documented Z-ID is the proposed IR-8 Cherenkov [8]). Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ vs $\sim 10\times$ flatter incoherent; crossover at $0.05\text{--}0.2 \text{ GeV}^2$ inside the window [41,42]); $e + \text{Pb}$ benchmark: 80–99% incoherent rejection [43]
Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), $^3\text{He} + t$ (15.79 MeV) [39]	A fragment could pass unnoticed	$p \rightarrow \text{OMD}$ ($R = 0.50$), $n \rightarrow \text{ZDC}$, $^3\text{He} \rightarrow \text{RP window}$ ($R = 0.75$) [51]: vetoable; inefficiency is second-order (BeAGLE study to quantify)
$^6\text{Li}^*$ 3.5629 MeV (0^+ , $T = 1$, $\Gamma = 8.2 \text{ eV}$) [39]	Particle-stable ($\alpha + d$ parity-forbidden, $\Gamma_{\alpha} \leq 6 \times 10^{-7} \text{ eV}$); γ -decays to a real $A = 6$, $Z = 3$ track	Isoscalar exchange cannot drive $T = 0 \rightarrow 1$; the boosted $0.1\text{--}0.4 \text{ GeV}$ γ lands mostly in the B0 calorimeter (photons $\geq 50 \text{ MeV}$, $5.5\text{--}20 \text{ mrad}$ [9]), tail in the ZDC. The 5.366 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is γ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha + d$ channel
Bethe–Heitler / QED Compton	Intact nucleus by construction	Calculable (t -dependence = charge FF^2 [11,12]); subtract and reuse as an RP-acceptance candle; the charge-quadrupole $\cos 2\varphi$ is $< 10^{-3}$ for $ t < 0.15 \text{ GeV}^2$ [12,17]
Beam-gas / halo; e' -side fakes	Accidental $R \approx 1$ tracks; π / pair-symmetric e' at high y	Non-colliding bunches, $\sim 30 \text{ ps}$ timing, vertex association; standard DIS e' ID plus (e' , recoil) kinematic consistency

Extraction systematics: the modulation is self-normalized within a single fill, so relative-luminosity systematics drop out; ϕ -acceptance holes are handled by the binned fit; the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$ is common with A_{zz} ; radiative corrections for tensor observables are uncalculated and flagged on every plot. At the interpretation level, a $\Delta\Delta$ -isobar admixture can mimic Δ at leading twist [34] and conventional double scattering generates the low- x b_1 [21–23]; the Q^2 lever arm and the ${}^6\text{Li}/d$ comparison are the discriminants.

The ${}^6\text{Li}$ null test: deformation-driven coherent modulations — the polarized-deuteron imaging signal [13] — are bounded on ${}^6\text{Li}$ by its anomalously small quadrupole moment, $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$ [35]. A sizable flat-in- $t \cos 2\phi$ on ${}^6\text{Li}$ cannot be a shape effect; the mechanisms entangled for the deuteron separate in the ${}^6\text{Li}/d$ comparison.

6 • Critical assumptions

#	ASSUMPTION	IMPACT IF WRONG	REPLACEMENT PATH
1	RP p_T cuts: 0.20 GeV is the documented top-rigidity envelope scale (${}^3\text{He}$ precedent [10]); 0.45 GeV is our derivation; no Li optics exist — proton-class divergence would give 0.5–0.7 GeV	Tag acceptance anywhere from 13.5% to negligible at IP6; the physics case shifts to IR-8	Li beam-optics parameters from EIC C-AD; the IR-8 secondary focus removes the cut entirely [10]
2	Scenario coherent fraction $f_0 = 0.04$ [$\times 2/\div 2$], parameterized in Bjorken x	Coherent yields scale linearly; δA as $1/\sqrt{f_0}$	Real diffractive model — open question #18; concrete ask: an IP-Glasma α -d run [13]
3	Bag-moment normalization of Δ ported to per-nucleon ${}^6\text{Li}$ with 1/3 dilution; x-shape ours	Overall Δ scale; literature brackets both directions (NPLQCD suppression [27] vs binding enhancement [50])	Lattice moment for a light nucleus; theory engagement (plans/04)
4	ϵ_{B0} scaling from the deuteron (two dials disagreeing by $\times 3$)	Deformation amplitude anywhere in the band; sign flip robust	The same IP-Glasma α -d calculation; band quoted on every plot
5	Event-by-event Z-ID in the RPs is NOT assumed (purity via $ t $ -shape fit)	If Z-ID exists purity improves; if the incoherent slope model is off, the fit systematic grows	#19 answer from the ePIC far-forward group; BeAGLE incoherent spectra
6	Linear $a_2(t)$ valid only for $ t < 0.2 \text{ GeV}^2$; form-factor zero at 0.31 GeV^2 excluded, not modeled	Upper tagged window carries unmodeled structure	Form-factor-level model of the dip region (deuteron analog in [13])
7	TOY structure functions; no target-mass terms; $g_2 = g_2^{WW}$	Rates to $\times 1.5$; shapes are scenarios	PDF-grid backends exist; digitized theory curves scheduled (plans/02)
8	All luminosity in the transverse-tensor fill; $P_{zz} = 0.6$ in-ring placeholder	N scales with the run-plan share; amplitude with P_{zz}	Run-plan bookkeeper supports splits; source demonstrations reach $\sim 95\%$ of maximal P_{zz} [37]
9	Impulse approximation; no FSI; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger $ t $; RC unquantified	Theory engagement (plans/04); the QED tail is calculable and doubles as calibration
10	Equal-rate population average for $\langle a_2 \rangle$; t_{min} neglected in the tag acceptance	One-sided $\sim 27\%$ (rate weighting) and $\sim 10\%$ (t_{min}) — inside the ϵ_{B0} / f_0 bands	Rate-weighted ensemble model (the IP-Glasma run supersedes both)

7 • Next steps

Near term: BeAGLE incoherent ${}^6\text{Li}$ spectra to turn the veto efficiency and $|t|$ -fit systematics into numbers (plans/02); VMC α -d/ α -t overlaps replacing the two-parameter radial forms (plans/04); pseudo-experiment closure with a 2-D (ϕ , ϕ_t) acceptance map (plans/03); HepMC3 output and an ePIC chain smoke test (plans/05) so these projections can be repeated at reconstructed level.

Theory asks: rerun the IP-Glasma polarized-deuteron setup [13] with an α -d cluster density — the single highest-leverage input, replacing both ϵ_{B0} dials and the f_0 band at once; a lattice Δ moment for a light

nucleus; the spin-3/2 structure-function basis for ${}^7\text{Li}$; tensor-observable radiative corrections; the sum-rule tie between transversity-GPD moments and measured tensor form factors [30].

Machine and detector asks: lithium beam-optics parameters (they set the IP6 p_T cut, assumption #1); an answer to #19 — can the RP AC-LGAD amplitude separate $Z = 3/2/1$ at identical rigidity and velocity?; and the IR-8 secondary focus with its proposed Z^2 Cherenkov [8], which lifts the tag toward $p_T \approx 0$ (${}^7\text{Li}$: 17.8% published; ${}^6\text{Li}$ not yet studied [10]) — the coherent- ${}^6\text{Li}$ program is one more physics case for the second interaction region. Off-momentum halo sets how close the pots sit, i.e. the cut itself.

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